Exploring Open-Source LLMs as Annotators for Bangla Language

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Abstract

Low-resource languages face significant challenges in natural language processing (NLP) due to insufficient linguistic data and resources. This scarcity hinders the development of accurate models for tasks like sentiment analysis and news classification. Large Language Models (LLMs) offer a potential solution as annotators for generating datasets in underrepresented languages. This study evaluates open-source LLMs as annotators for Bangla, a low-resource language. We assess Llama 3.2 (3B and 1B) and Gemma 2 (2B) on classification tasks including sentiment analysis, news classification, and authenticity detection across multiple Bangla datasets. Our findings reveal that while LLMs excel in annotation tasks for high-resource languages like English, they still struggle when applied to low-resourced language as Bangla. The performance of these models varies across different classification tasks, highlighting the complexities of annotating low-resource language datasets.

1 Background

Machine learning tasks such as supervised learning, heavily rely on labeled data, making data annotation a fundamental step. Data annotation involves labeling or tagging data to make it understandable for machines. This process is critical for both building and evaluating machine learning models (Tan et al., 2024). Consequently, the quality and reliability of annotations are important to developing effective ML models (Jadhav et al., 2024). However, curating high-quality data annotations is often expensive and time-consuming, especially for low-resource languages. In this context, With the increasing prevalence of large language models (LLMs), recent studies have explored their potential as data annotators for machine learning tasks such as text classification and sentiment analysis (Sun et al., 2023; Nasution and Onan, 2024; Zhu et al., 2023; Kuzman and Ljubešić, 2023; Gao et al., 2023). However, despite these advancements, there is a notable gap in leveraging LLMs for resource-constrained languages, such as Bangla. To date, no prior work has explored the performance of LLMs as annotators for Bangla datasets. This raises a critical question: *how well can LLMs perform annotation tasks in low-resource languages like Bangla, where linguistic and cultural nuances are unique and challenging?*

To bridge this gap, our study examines the role of LLMs as annotators for Bangla datasets, exploring their potential to replace traditional crowd-sourced annotation methods. Our key contributions are as follows:

- We assess the effectiveness of LLMs in annotating Bangla datasets, analyzing their performance based on accuracy
- We provide insights into the practicality of using LLMs as annotation alternatives in low-resource language (Bangla) settings.

This paper is structured as follows: section 1 provides background information. section 2 presents the methodology, detailing the datasets and models employed in the experiment. section 3 discusses the results of the evaluation. section 4 offers a discussion of the findings. section 5 outlines the limitations of the study. Finally, section 6 concludes the paper.

2 Methodology

We investigate the distinctions between LLM-generated and human-generated annotations for Bangla language datasets using a comparative methodology. Our research focuses on two tasks using relevant Bangla datasets: 1)

Dataset	Prompt	Llama 3.2 (3B)	Llama 3.2 (1B)	Gemma 2 (2B)
BanglaFoodReview	ZS	0.80	0.44	0.81
	FS	0.78	0.54	0.85
	CoT	0.78	0.55	0.86
SentNoB	ZS	0.25	0.27	0.31
	FS	0.26	0.29	0.40
	CoT	0.25	0.32	0.41
RevBangla	ZS	0.89	0.72	0.60
	FS	0.88	0.75	0.50
	CoT	0.87	0.77	0.70
BanFakeNews	ZS	0.57	0.72	-
	FS	0.66	0.85	-
	CoT	0.57	0.21	-

Table 1: LLM models comparison across different tasks. Prompt: Different prompting techniques Used; ZS: Zero-shot; FS: Few-shot; CoT: Chain-of-thought.

BanglaFoodReview (Junaid et al., 2022) - classifies sentiments of Bangla Food Review into positive and negative categories, 2) RevBangla (Sunny et al., 2024) - classifies sentiments of Bangla customer reviews into positive and negative categories, 3) SentNoB (Islam et al., 2021) - classifies sentiments of noisy bangla comments from different domains into positive, negative and neutral categories, 4) BanFakeNews (Hossain et al., 2020) - classifies news into authentic and fake categories.

2.1 LLMs

In our annotation experiment, we have evaluated the performance of LLMs for the Bangla language using three prompting techniques: zero-shot, few-shot learning and Chain-of-Thought. We tested on open-source models (Llama 3.2 3B, Llama 1B and Gemma 2 2B). The performance of each LLM under all the prompting strategies is summarized in Table 1.

3 Result

This section demonstrates a detailed overview of the experiments conducted for the annotation of 4 datasets utilizing three large language models (LLMs). All the results are summarized in Table 1. We have considered only one metrics: accuracy following similar works (Gilardi et al., 2023; Jadhav et al., 2024; He et al., 2023; Sun et al., 2023). The table facilitates a comprehensive evaluation by highlighting key outcomes, enabling a thorough understanding of how each model performs across the different annotation tasks and prompting methods.

4 Discussion

Our findings reveal that current Large Language Models (LLMs) exhibit limitations in effectively handling Bangla-language text annotation tasks. As shown in Table 1, the performance of Llama 3.2 (1B and 3B) and Gemma 2 (2B) across diverse datasets, including BanglaFoodReview, SentNoB, RevBangla, and BanFakeNews, indicates that LLMs still face challenges in achieving reliability and accuracy comparable to human manual annotations.

For sentiment classification tasks, such as RevBangla, Llama 3.2(3B) model achieved the highest performance across all prompting techniques, demonstrating its relative advantage over other models.However, for tasks requiring nuanced understanding, like SentNoB, all models struggled, particularly in zero-shot (ZS) prompting settings.As the SentNoB dataset classifies sentiments of noisy Bangla comments from different domains into positive, negative and neutral categories which is not binary classification, all the prompting settings fall behind in performance. It becomes difficult to reason over the sentiments and classify them among more than 2 classes .Chain of Thought (COT) prompting yielded slightly better results, particularly for Gemma 2 in case of BanglaFoodReview dataset,whereas Llama 3.2 (3B) model is giving better accuracy in case of RevBangla dataset.

In the BanFakeNews dataset, which involves reasoning over contents of news creates complexity

for fake news detection, performance was disappointing significantly by Gemma2 (2B) model. In the process of generating annotations for this dataset, most of the time Gemma2(2B) model was generating noisy labels for inputs. Because the contents of the news in this dataset were quite lengthy and it was difficult for this model to reason over and label. While Llama 3.2 (1B) demonstrated reasonable accuracy in few-shot prompting, other configurations, such as CoT prompting for both Llama 3.2 (1B) and Gemma 2 (2B), failed to deliver consistent results.

These observations highlight the challenges of adapting LLMs to resource-constrained languages like Bangla, where linguistic complexity and lack of high-quality training data can impact model performance.This underscores the need for further research into optimizing LLMs for Bangla language to bridge the gap between LLM and human-level annotation quality.

5 Limitations

This study is limited in several ways. First, we were unable to explore the performance of closed-source models due to access constraints. Many closed-source LLMs which might have performed better were not included in our evaluation. Second, even for open-source models, we were restricted to using models such as Llama and Gemma with fewer parameters due to computational resource constraints. Larger models with more parameters could potentially provide improved performance, but their evaluation was not feasible in this work. Lastly, while we focused on a few specific classification tasks, there is a broader range of tasks and datasets that could be examined to provide a more comprehensive understanding of LLMs' capabilities as annotators for low-resource languages like Bangla. Future work can address these limitations by incorporating a wider range of models, tasks, and datasets to strengthen the findings.

6 Conclusion

Our study highlights that while LLMs such as Llama 3.2 and Gemma exhibit potential, they are not yet reliable annotators for low-resource languages like Bangla, especially for complex classification tasks. These results emphasize the need for further advancements in LLMs to establish them as viable alternatives to human annotations. This research demonstrates the importance of developing more robust models that are specifically tailored to the linguistic nuances and unique challenges of low-resource languages like Bangla. By addressing these limitations, future advancements could significantly reduce reliance on manual annotation processes. Overall, our findings reveal the considerable challenges in leveraging LLMs for annotation tasks in low-resourced linguistic settings, emphasizing the necessity for continued research and targeted model development to meet the needs of such languages.

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